Subject-Level Calibration Heterogeneity is Stable Across LLM Families and Scales: A Statistical Audit of MMLU

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Abstract

Aggregate calibration metrics obscure a critical risk: LLMs are most overconfident in the specialized domains—legal reasoning, medical science, technical STEM—where users are least equipped to catch errors independently. We develop a statistical audit framework—a multilevel decomposition of question-level calibration gap variance, isotonic reliability curves per subject, empirical Bayes shrinkage, and Benjamini–Hochberg FDR correction—and applied it to MMLU using seven instruction-tuned models from two families across six parameter scales. The framework recovers a coherent contrast in Qwen-2.5-{0.5B, 1.5B, 3B, 7B, 14B}-Instruct. Three findings emerge. (i) Among the six larger models, subject-level ECE rankings are highly correlated across all 15 model pairs (Spearman $\rho \in [0.78, 0.94]$), with cross-family agreement at matched capability matching the strongest within-family pairs. (ii) The same subjects are robustly worst-calibrated (college-level quantitative STEM, niche factual recall, structured ethical/legal reasoning) and robustly best-calibrated (high-school-level Social Sciences and applied/factual subjects), revealing specialization level as a meaningful axis of calibration quality. (iii) Within the Qwen family at matched precision, both between-subject ICC and the entropy coefficient grow log-linearly across the 1.5B→3B→7B→14B scaling sweep, consistent with prior literature on RLHF-induced confidence saturation (**??**). The smallest model in our set (Qwen-0.5B) shows extreme positional bias and a qualitatively different miscalibration pattern, suggesting a lower bound below which the elicitation protocol breaks down. The stability of audit findings across the larger model space supplies the structural justification for subject-aware recalibration as in **?**, and the audit itself supplies a transferable diagnostic for new models.

1. Introduction

When users consult an LLM for legal guidance, medical information, or technical scientific questions, they typically lack the domain expertise to evaluate the answer independently—placing full weight on the model's expressed confidence as a reliability signal. Systematic overconfidence in these domains therefore creates a predictable, population-scale harm: users acting on high-confidence wrong answers have no reliable way to detect the error. Our findings make this concrete: across seven models from two families spanning a 28× scale range, the most severely miscalibrated subjects are stably the professional and technical domains—`professional_law`, `virology`, `college_chemistry`, `college_mathematics`—precisely where reliable confidence signals matter most. Yet the standard calibration literature reports only aggregate scalars—ECE (**?**), NLL, or related—that are silent about where miscalibration concentrates. We ask three questions:

1. **Where is calibration good and bad within a benchmark?**
2. **Are these patterns stable across models?**
3. **What mechanisms produce the observed heterogeneity?**

Our contribution is a statistical audit framework—multilevel decomposition, FDR correction, isotonic curves with empirical Bayes shrinkage—that produces direct answers to these questions, and an application of the framework to seven instruction-tuned LLMs on MMLU (**?**). The framework is diagnostic rather than corrective: **?** demonstrates that subject-aware temperature scaling improves on global recalibration; we ask whether the structural assumption underlying that scoping decision (that miscalibration patterns are stable enough to be worth scoping over) is empirically justified.

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Prior work has studied LLM calibration mostly at the aggregate level. **?** showed that token-level logprobs carry a meaningful calibration signal on multiple-choice tasks and that instruction tuning tends to degrade it; **?** noted calibration degrades on tasks requiring professional knowledge but did not investigate subject-level structure formally. **?** and **?** document RLHF-induced confidence saturation at the aggregate level. We bring a formal statistical audit to the within-benchmark heterogeneity these works describe only in passing.

2. Method

Confidence elicitation. For each MMLU question we extract token-level log probabilities over $\{A, B, C, D\}$, softmax-renormalize, and define c_i as the maximum renormalized probability. The signed calibration gap $\mathrm{gap}_i = c_i - y_i$ (where $y_i \in \{0, 1\}$ is correctness) is our primary outcome. GPT-4o-mini is queried via the OpenAI Batch API with `top_logprobs=20`; the open-weight models are run on a Colab GPU with the assistant turn pre-filled by "The answer is" to score the next-token position. The 3B, 7B, 8B, and 14B models use 4-bit quantization (BitsAndBytes; **?**); the 1.5B model is run twice (4-bit and fp16) so the precision effect can be examined directly, with the fp16 version used as the canonical 1.5B in the main analysis. Test-set accuracies: 0.749, 0.606, 0.405, 0.571, 0.658, 0.601, 0.685, 0.757 for GPT, Llama, Qwen-{0.5B, 1.5B fp16, 3B, 7B, 14B}. Qwen-0.5B exhibits extreme positional bias (predicting D 59% of the time vs. A 5.6%), a confound we discuss where it affects results.

Multilevel decomposition. We fit a three-level mixed-effects model with questions nested in subjects nested in domains:

$$\mathrm{gap}_{ijk} = \mu + \mathbf{x}_{ijk}^\top \boldsymbol{\beta} + u_k + v_{jk} + \varepsilon_{ijk}, \qquad (1)$$

where u_k is a domain random intercept, v_{jk} a subject-within-domain intercept, and $\mathbf{x}_{ijk}$ standardized question-level covariates: word count, max option length, negation indicator, and Shannon entropy of the A/B/C/D distribution ($H_i = -\sum_k p_{ik} \log p_{ik}$ in nats). Variance components are estimated by REML; the ICC decomposition reports the share of total variance at each level.

Calibration curves and ECE. For each subject we fit isotonic regression (**?**) and an unconstrained Gaussian kernel smoother ($\sigma = 1.5$ bins) on (c_i, y_i), and compute ECE as the 20-bin weighted absolute deviation between the curve and the diagonal.

Multiple testing. For each of 57 subjects we test H_0: mean calibration gap $= 0$ via a one-sample t-test on question-level gaps, then apply Benjamini–Hochberg FDR correction at $\alpha = 0.05$.

3. Results

3.1. Variance decomposition (RQ1)

Table 1 reports the ICC decomposition. Across all seven models, the bulk of calibration-gap variance lies at the question level (92.1–98.3%) and domain-level variance collapses to zero. Between-subject ICCs span nearly a factor of five, from 1.7% (Qwen-1.5B) to 7.9% (GPT-4o-mini). Within the Qwen family at matched precision (1.5B fp16, 3B 4-bit, 7B 4-bit, 14B 4-bit), ICC grows log-linearly: $0.017 \to 0.051 \to 0.049 \to 0.059$.

Table 1. Variance decomposition of the calibration gap. Domain-level variance is estimated at zero for every model; $\mathrm{ICC}_{\text{subject}}$ is the share of total variance at the subject level.

Model	$\sigma^2_{\text{subject}}$	σ^2_ε	$\mathrm{ICC}_{\text{subject}}$
GPT-4o-mini	0.013	0.150	0.079
Llama-3-8B-Instruct	0.004	0.190	0.020
Qwen-2.5-0.5B-Instruct	0.004	0.224	0.018
Qwen-2.5-1.5B-Instruct	0.003	0.197	0.017
Qwen-2.5-3B-Instruct	0.011	0.203	0.051
Qwen-2.5-7B-Instruct	0.009	0.177	0.049
Qwen-2.5-14B-Instruct	0.010	0.156	0.059

3.2. Where calibration succeeds vs. fails (RQ1)

The same subject clusters appear at both extremes of the ECE distribution across the six larger models (Qwen-0.5B is excluded from the cluster analysis due to its positional-bias confound; Section 3.3). Four subjects are top-10 most miscalibrated for every model: `virology`, `college_chemistry`, `college_physics`, `professional_law`. Top-15 expansion adds `college_mathematics`, `econometrics`, `high_school_physics`, `machine_learning`, and `moral_scenarios`. These cluster into three semantic groups: college-level quantitative STEM, niche factual recall, and structured ethical/legal reasoning—precisely the high-stakes domains where systematic overconfidence creates the largest harm potential.

Symmetrically, two subjects are bottom-5 best-calibrated for every larger model (`high_school_government_and_politics`, `high_school_psychology`); bottom-10 adds `high_school_us_history`, `marketing`, and `miscellaneous`. Two patterns:

- **Specialization level matters.** MMLU's high-school-level subjects systematically show smaller calibration gaps than their college-level counterparts (compare high_school_psychology to professional and college subjects in similar areas).

- **Domain category matters.** Best-calibrated subjects are dominated by Social Sciences and the "Other" (applied/factual) MMLU category; no STEM subject appears in the bottom-10 intersection across models.

Figure 1 contrasts top-5 most-miscalibrated and bottom-5 best-calibrated reliability curves for GPT-4o-mini. Curves are masked to the 1st–99th percentile of observed confidence per subject to avoid displaying isotonic extrapolation in regions with no data. The two panels reveal the heterogeneity that aggregate ECE conceals: in a single model on a single benchmark, some subjects sit on the calibration diagonal while others fall sharply below it.

3.3. Cross-model agreement (RQ2)

Figure 2 reports the 7×7 Spearman rank correlation matrix of subject-level ECE. Among the six larger models, all 15 pairwise correlations fall in $[0.78, 0.94]$, with the strongest pair Qwen-7B vs Qwen-14B at 0.94 and the cross-family GPT-4o-mini vs Qwen-14B and GPT-4o-mini vs Qwen-7B both at 0.92. Subject-level miscalibration patterns are therefore consistent across model families and scales above approximately 1B parameters.

Qwen-0.5B is the conspicuous exception: its correlations with the other six models fall in $[0.16, 0.22]$, with bootstrap 95% CIs that include zero. The underlying cause is extreme positional bias—Qwen-0.5B predicts D on 59% of questions and A on only 5.6%, vs. approximately uniform 25% for the larger models. The 0.5B's miscalibration is therefore dominated by a spurious preference for the D position rather than genuine knowledge gaps, producing a qualitatively distinct ranking of which subjects appear miscalibrated. We interpret this as evidence that, below approximately 1B parameters, positional bias dominates the calibration signal in our elicitation protocol.

3.4. Mechanism: predictors and within-family scale (RQ3)

Table 2 reports the fixed effects from the multilevel model. Intercepts confirm large mean overconfidence in every model (0.179–0.311). Within the Qwen family, the intercept peaks at 3B (0.311) where strong confidence priors have been acquired ahead of capability gains, and declines at 7B and 14B, consistent with accuracy gains partially offsetting the underlying confidence inflation.

Entropy: a sign progression with scale. The most informative predictor is entropy. A naïve prediction is that high option-distribution entropy reduces the calibration gap, since top-1 confidence is mechanically lower when probability mass spreads. Yet within the Qwen family at matched precision the coefficient progresses log-linearly with scale: -0.004 (n.s.) at 1.5B, $+0.006$ (n.s.) at 3B, $+0.029$ ($p < 0.001$) at 7B, $+0.040$ ($p < 0.001$) at 14B. Figure 4 traces the mechanism by partitioning each model's questions into entropy quartiles and plotting mean confidence (solid) vs. mean accuracy (dashed). At 1.5B confidence and accuracy fall in roughly parallel as entropy rises (gap $0.089 \rightarrow 0.134$); at 7B the model holds confidence near 1.000 across the first three quartiles while accuracy drops to 0.576; at 14B this saturation extends further, with confidence still at 1.000 even when accuracy has fallen to 0.709.

Table 2. Fixed effects from the three-level mixed model. Continuous covariates standardized. Significance: $^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$, $^{\dagger}p < 0.10$.

Predictor	GPT	Llama	Q-0.5B	Q-1.5B	Q-3B	Q-
Intercept	0.196^{***}	0.197^{***}	0.209^{***}	0.179^{***}	0.311^{***}	0.233
Word count	0.031^{***}	-0.005	-0.013^{*}	0.003	0.010	0.01
Max opt len	0.001	-0.001	-0.028^{***}	0.005	$-0.008^{\dagger}$	-0.0
Negation	-0.001	0.007	0.016	-0.010	-0.014	-0.0
Entropy	0.016^{***}	$0.008^{\dagger}$	-0.063^{***}	-0.004	0.006	0.029

This pattern aligns with prior aggregate findings on RLHF and confidence (**???**): more capable, more aggressively-aligned models maintain confidence near saturation even when the option distribution flattens. We are cautious about extending this further: the within-family scaling comparison is observational rather than controlled, and Qwen-0.5B's strongly negative entropy coefficient (-0.063, $p < 0.001$) is best read as an artifact of its positional bias rather than a fourth scale point. The marginal pattern is consistent with the literature; a stronger causal claim would require additional small or base models.

Figure 5 summarizes the within-Qwen scaling on the four matched-precision checkpoints (1.5B fp16, 3B 4-bit, 7B 4-bit, 14B 4-bit): ICC and the entropy coefficient grow log-linearly; the mean-overconfidence intercept is non-monotonic.

3.5. FDR-corrected rankings and sensitivity

At $\alpha = 0.05$: GPT-4o-mini, Llama, Qwen-3B, Qwen-7B, and Qwen-14B reject all 57 subjects; Qwen-1.5B (fp16) rejects 56/57, and Qwen-0.5B rejects 55/57. The substantive finding is the magnitude of heterogeneity: per-model gap ranges span 5.4–13.0× from the best-calibrated to the worst-calibrated subject in the larger six models.

Sensitivity-analysis conclusions extend uniformly: ECE rankings are stable across bin counts (Spearman ≥ 0.97 vs. 20-bin baseline), proper-scoring NLCS rankings track ECE rankings (Spearman ≥ 0.79), and two-level vs. three-level

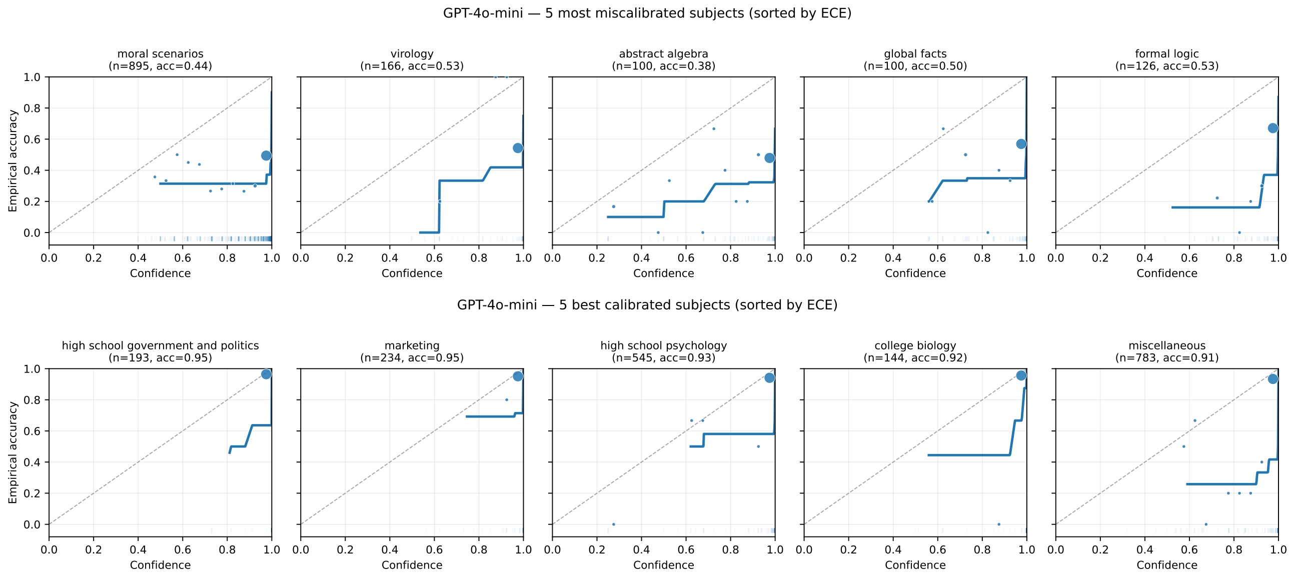

Figure 1. GPT-4o-mini reliability curves. **Top**: five most miscalibrated subjects by ECE. **Bottom**: five best-calibrated subjects. Same conventions in both: solid line is isotonic regression, plotted only on observed support; markers are 20-bin empirical accuracy sized by bin count; dashed line is the calibration diagonal.

ICC estimates are numerically identical (consistent with $\sigma^2_{\text{domain}} = 0$).

4. Discussion

What the audit recovers. Aggregate ECE understates structured heterogeneity that is real (FDR-rejected, robust to bin choice, robust to proper-scoring metrics), task-driven (correlated across families and scales above 1B parameters), and aligned with intuitive content axes (specialization level, domain category). The robust well-calibrated set (high-school-level Social Sciences and applied subjects) and the robust poorly-calibrated set (college-level quantitative STEM, niche factual recall, structured reasoning) appear in every model larger than 1B that we examined. Within the Qwen family at matched precision, both the between-subject ICC and the entropy coefficient grow log-linearly across the 1.5B→3B→7B→14B scaling sweep, providing within-family corroboration of the cross-family pattern.

Implications for recalibration. Stable cross-model targets provide the structural justification for choosing subject-aware recalibration scoping over global recalibration. We do not re-fit the corrections themselves (those would still be model-specific): **?** demonstrates the operational arm directly. Our framework supplies the diagnostic upstream: when miscalibration patterns are this stable across very different training pipelines, scoping a correction at the subject level is the structurally appropriate choice. The audit findings also provide a transferable diagnostic: the robustly worst-calibrated subjects we identify can flag miscalibration on new models before deployment.

The audit as an accountability instrument. The cross-model stability of our findings has a direct governance implication: because subject-level miscalibration patterns are consistent across model families above approximately 1B parameters, the audit protocol is transferable without re-establishing baseline comparisons. The robustly worst-calibrated subjects—`professional_law`, `virology`, `college_chemistry`, `moral_scenarios`—constitute a natural adversarial test set for pre-deployment evaluation in high-stakes contexts. Deployers and regulators can apply this protocol as a targeted accountability checkpoint, surfacing structured overconfidence risks that aggregate ECE would not reveal and that model cards reporting only top-line metrics would obscure. The within-family scaling pattern further suggests that as model capability grows, the concentration of miscalibration into specific subject clusters becomes more pronounced, not less—making subject-aware diagnostics increasingly important at the largest deployed scales.

Limitations. The within-family scale comparison rests on four checkpoints (Qwen 1.5B, 3B, 7B, 14B); replication with additional scale points (Qwen-2.5-32B and beyond) and additional small or base-model checkpoints would test whether the trend extends. Cross-family generalization would benefit from a third family (Mistral-7B or Gemma-2-9B) at matched scale. 4-bit quantization affects five of

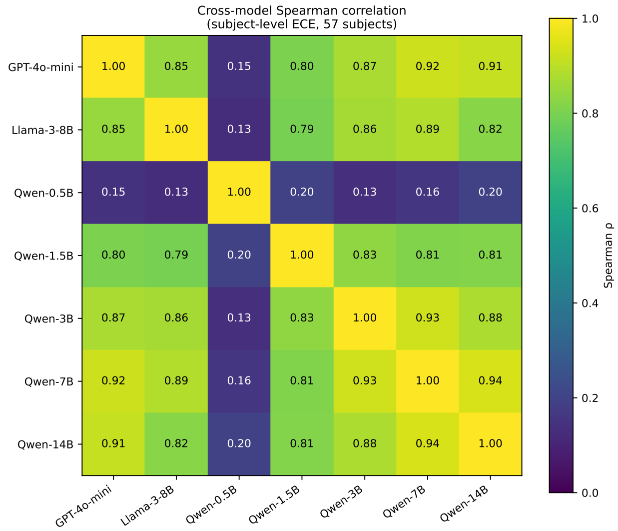

Figure 2. Spearman rank correlation matrix of subject-level ECE across the seven models (57 common subjects). Among the six larger models, all 15 pairwise correlations fall in $[0.78, 0.94]$. Qwen-0.5B's correlations with the others are $[0.16, 0.22]$, reflecting a qualitatively different miscalibration pattern driven by extreme positional bias.

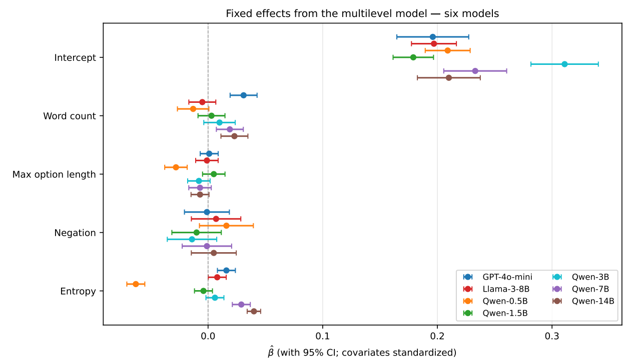

Figure 3. Fixed effect estimates with 95% confidence intervals across the seven models. The entropy coefficient progression—from strongly negative at Qwen-0.5B through near-zero at Qwen-1.5B to strongly positive at the larger Qwens—is the most informative pattern.

the seven models studied and may modestly attenuate confidence; we conducted a precision-effect comparison at 1.5B (re-running fp16) but not at larger scales. Qwen-0.5B's extreme positional bias makes it unusable as a fourth scaling point but is itself an informative finding about the lower bound of the elicitation protocol. Findings are specific to MMLU's four-choice format and do not directly extend to open-ended generation.

5. Conclusion

Subject-level calibration heterogeneity in LLMs is real, large, and stable across model families above approximately 1B parameters. Applied/factual high-school subjects are reliably well-calibrated across all models we examined; college-level technical content, niche factual recall, and structured ethical/legal reasoning are reliably poorly calibrated. Subject-level ECE rankings are highly correlated across all 15 pairs of larger models ($\rho \in [0.78, 0.94]$), including cross-family pairs at matched capability, establishing these patterns as properties of the task rather than any particular training pipeline. Within the Qwen family at matched precision, both ICC and the entropy coefficient grow log-linearly across the 1.5B→3B→7B→14B scaling sweep—an observational pattern consistent with RLHF-induced confidence saturation. The audit supplies the structural justification for subject-aware recalibration scoping, a transferable pre-deployment diagnostic for new models, and a concrete accountability instrument for AI deployment in the legal, medical, and scientific domains where calibration failures carry the greatest societal cost.

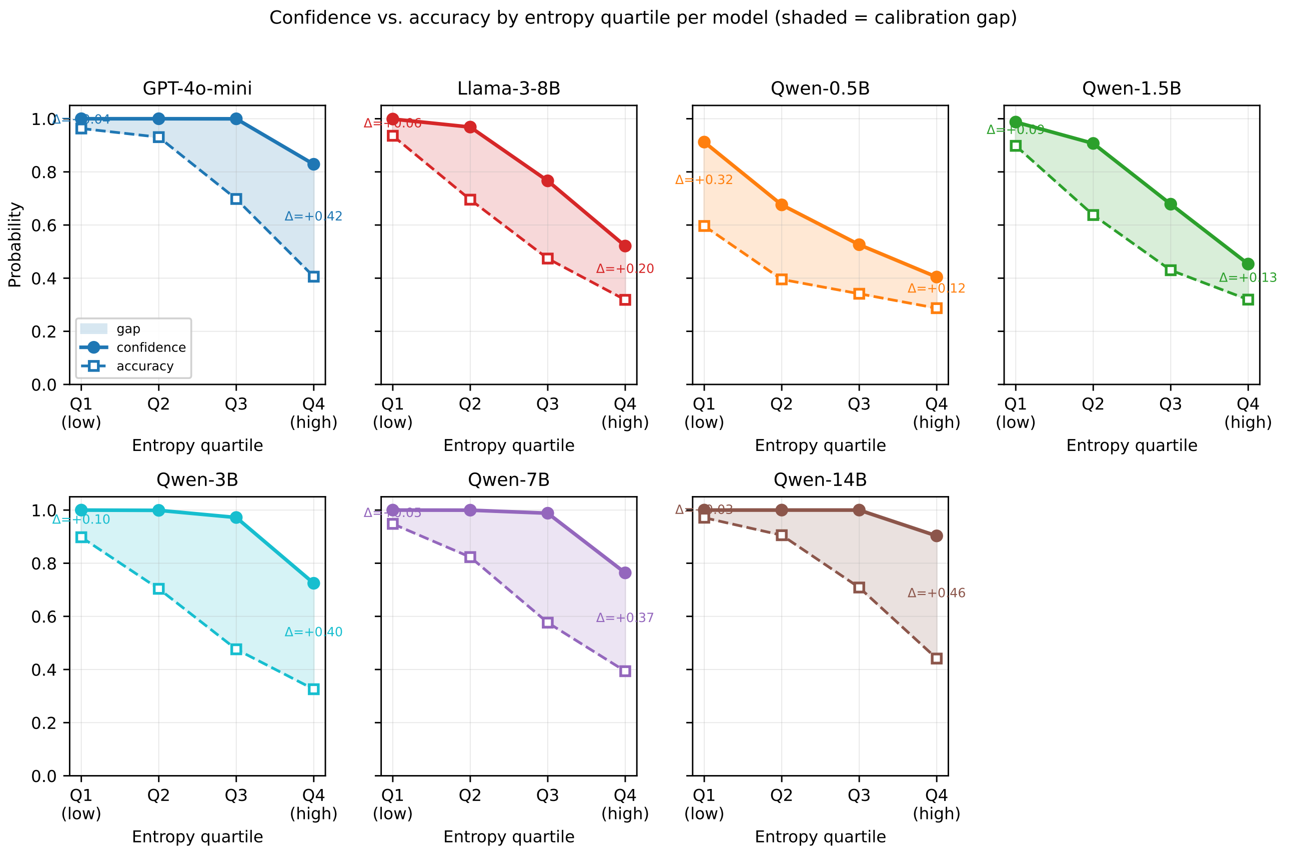

Figure 4. Confidence (solid) and accuracy (dashed) by entropy quartile per model. Shaded area is the calibration gap. The within-Qwen progression is systematic: at 1.5B the curves fall in parallel; at 7B and 14B confidence remains near 1.0 across most quartiles even as accuracy drops sharply.

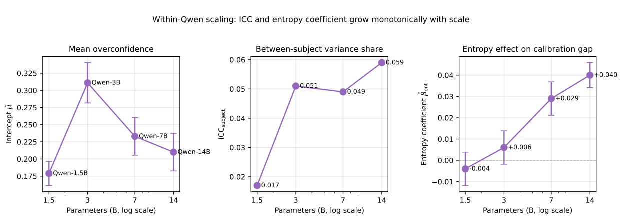

Figure 5. Within-Qwen scaling on the four matched-precision checkpoints (1.5B fp16, 3B 4-bit, 7B 4-bit, 14B 4-bit). Both ICC and the entropy coefficient grow log-linearly with scale.

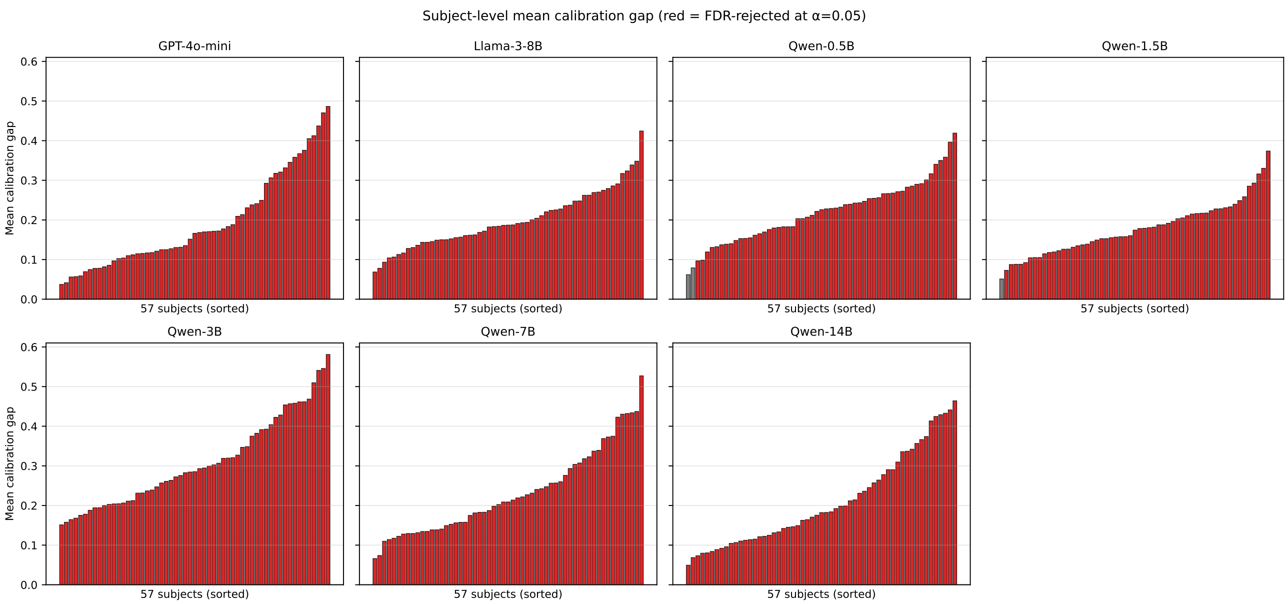

Figure 6. Mean calibration gap per subject, sorted ascending, for all seven models. Red bars are FDR-rejected at $\alpha = 0.05$; gray bars are not. Five of seven models reject all 57 subjects.